

Parkinson's Pattern Profiling: A Statistical and Visual Modeling Study

Ananya Chepuri, Thomas Stranick

ac2857@scarletmail.rutgers.edu, tcs121@scarletmail.rutgers.edu

Abstract—This study presents a lightweight, interpretable approach to Parkinson's Disease (PD) detection using spiral handwriting images from the HandPD dataset. The project integrates handcrafted features, CycleGAN-based augmentation, and Vision Transformers (ViT). Key features like radial deviation, stroke length, curvature variance, and stroke density were extracted using OpenCV. Principal Component Analysis (PCA), t-tests, and shallow classifiers (Logistic Regression, KNN, Decision Tree) were used for statistical modeling. Results demonstrate strong classification accuracy exceeding 80% without deep feature fusion. ViT-based models, both pre-trained and custom, showed accuracy up to 91% with CycleGAN augmentation.

I. INTRODUCTION

Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting over 10 million people worldwide, including 1 million in the U.S. [1]. It primarily impairs motor function, causing tremors, rigidity, and movement difficulties. The diagnosis typically relies on clinical observation, patient history, blood tests, and imaging techniques to rule out other conditions [2]. One of the early observable symptoms of PD is hand tremor, which affects a patient's ability to perform fine motor tasks like handwriting. The HandPD dataset [3] contains spiral and meander tracing images from PD patients and healthy individuals, capturing these tremor-related motor deficits. In earlier studies, image processing techniques were used to extract features such as the Root Mean Squared Error (RMS) between the template and the traced drawings. These features were then used in classification models like Naïve Bayes, Optimum-Path Forest, and Support Vector Machines (SVMs) [4].

Using this dataset, we would like to see if we can expand on the work done in identification of Parkinsons by image recognition. With interpretable handcrafted features for traditional machine learning methods, and deep learning representations from Vision Transformers, we would like to detect Parkinson's Disease from handwriting samples. Our central research question is:

Do Vision Transformers (ViTs) perform better than traditional methods such as logistic regression, decision tree, and K nearest neighbors in predicting whether a given patient has Parkinson's based on traced templates that observe tremors?

II. DATASET AND CYCLEGAN

The HandPD [3] dataset was created to observe tremors in Parkinson's patients, and originally contains 72 control and 296 patient handwriting samples for both spiral and meander classes. For statistical models to perform at their

best, they need a large diverse sample size. To improve data diversity, mitigate class imbalance, and grow the sample size, four CycleGAN models were used to generate control and patient images for both types of handwriting samples. A CycleGAN model was chosen because the generated output can be controlled and trained to resemble both the input and output domains.

A CycleGAN is a neural network that aims to transform an input image to look more like another. The classic example being transforming a horse image into a zebra [5]. In our case, we would like to transform control images to resemble Parkinson's patients and vice versa. The way it accomplishes this is through two traditional GANs, each containing a generator network and a discriminator. The first generator's job is to generate an image that resembles the desired output, without completely changing the input. The discriminator can then determine if it looks enough like the desired output image. The second generator is used to make sure that the fake image can be turned back to the original image.

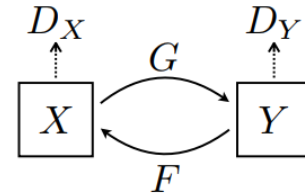


Fig. 1. CycleGAN Model Diagram from [5]

In our case, we implemented CycleGAN to generate 736 synthetic handwriting images by using original control and patient images as the inputs. We manually reviewed and cleaned several spiral and meander samples to improve training quality:

- Removed outlier samples in each group
- Prioritized samples in control where the spiral was cleanly traced and free from irregular distortions

Four separate generators were trained for each combination of group and class:

- Control $\rightarrow$ Patient (Spiral)
- Patient $\rightarrow$ Control (Spiral)
- Control $\rightarrow$ Patient (Meander)
- Patient $\rightarrow$ Control (Meander)

Each generator was trained for 200 epochs using a ResNet-based architecture with tuned cycle and identity loss weights.

We added random jitter and noise to the input to improve variability in the output, referencing techniques from the official TensorFlow CycleGAN tutorial [5].

After training the generators, we could feed all of our patient and control images into it and create new samples. After collecting the output images, we observed:

- Control image generation was highly effective, smoothing jagged lines and removing traces that were too far from the outline, producing realistic samples (Fig. 2)
- Patient image generation was less reliable, often introducing inconsistent or noisy blue lines (artifacts), but more resembling a patient with tremors (Fig. 3)

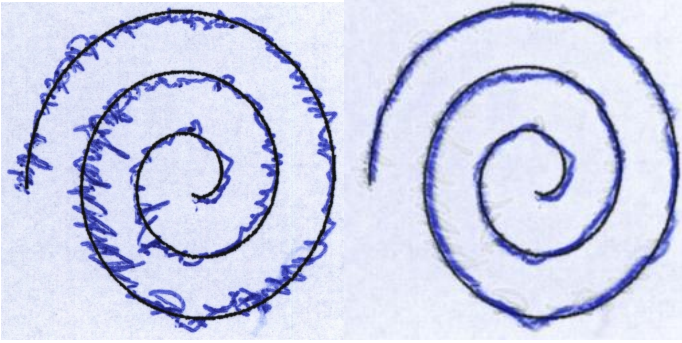


Fig. 2. CycleGan generated images Patient to Control.

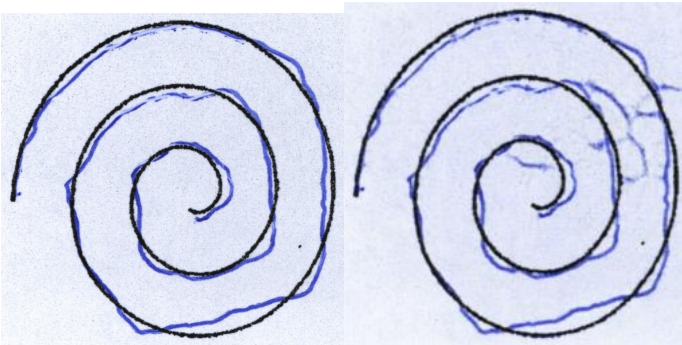


Fig. 3. CycleGan generated images Control to Patient.

With our new set of HandPD and CycleGAN generated images combined, we could find handcrafted features to show that they can differentiate between patient and control, and utilize them in some traditional machine learning techniques.

III. FEATURE ENGINEERING

Handwriting images were processed to extract interpretable motor features using OpenCV and NumPy. The following four features were engineered from each spiral image:

- **Radial Deviation:** Measures the variation in contour distances from the spiral's center, indicating tremor intensity.
- **Stroke Length:** Total contour arc length, reflecting stroke smoothness and completeness.
- **Curvature Variance:** Quantifies irregularity in stroke direction, capturing motion instability.

- **Stroke Density:** Ratio of inked pixels post-binarization, reflecting stroke compactness typical in PD handwriting.

Each of these features was computed for all real and synthetic images. The feature distributions were analyzed across the Control and PD classes using both visual and statistical tools:

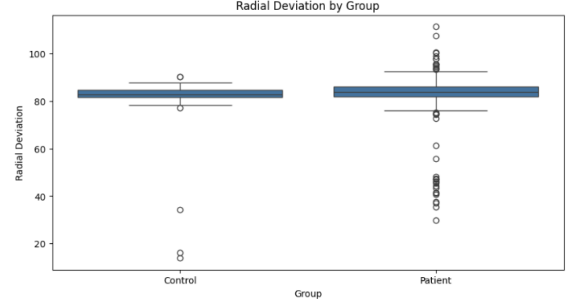


Fig. 4. Boxplot showing higher radial deviation in PD subjects.

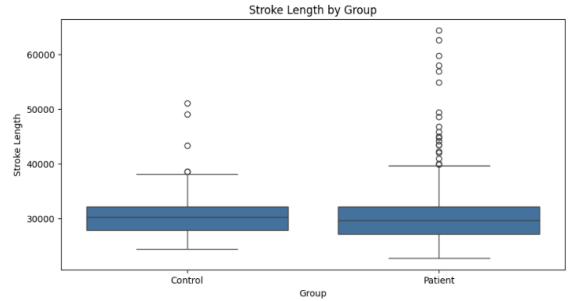


Fig. 5. Boxplot showing stroke length in PD subjects.

IV. PCA AND STATISTICAL TESTING

To explore the underlying structure of handwriting characteristics and assess their potential to distinguish between patients with Parkinson's disease (PD) and control subjects, Principal Component Analysis (PCA) was used. PCA is a dimensionality reduction technique that transforms a set of possibly correlated variables into a set of linearly uncorrelated variables called principal components. This transformation is achieved in such a way that the first principal component accounts for the largest possible variance in the data, with each subsequent component accounting for the remaining variance under the constraint of being orthogonal to the preceding components.

In this study, PCA was applied to four hand-crafted features extracted from handwriting samples. Radial Deviation (RD), Stroke Length (SL), Curvature (C), and Stroke Density (SD).

Visualization of PCA results demonstrated a discernible separation between PD patients and control subjects, particularly along the axes corresponding to RD and SD.

To statistically validate the differences observed in handwriting characteristics between PD patients and control sub-

jects, independent two-sample t-tests were performed for each characteristic. The results are summarized in the table below.

TABLE I
TWO-SAMPLE T-TEST RESULTS

Feature	p-value	Significance
(PCA1)Radial Deviation	0.0138	Significant ($p < 0.05$)
Stroke Length	0.5951	Not Significant
Curvature	0.0999	Borderline
(PCA2)Stroke Density	0.0444	Significant ($p < 0.05$)

The results of the t-test indicate that the radial deviation and the density of the stroke exhibit statistically significant differences between PD patients and control subjects, reinforcing their importance as discriminatory features. Curvature shows a marginal significance, suggesting a potential, albeit weaker, discriminative power. Stroke Length, however, does not demonstrate a statistically significant difference, implying limited utility in distinguishing between the two groups.

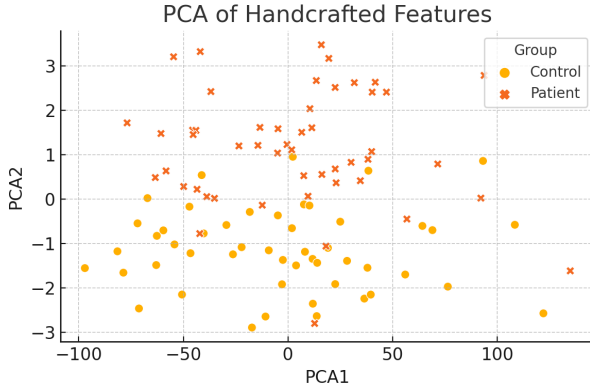


Fig. 6. PCA of handcrafted features

V. MODELLING AND CLASSIFICATION

To evaluate the discriminative power of the extracted handwriting features—Radial Deviation (RD), Stroke Length (SL), Curvature (C), and Stroke Density (SD)—we implemented three traditional machine learning models: Logistic Regression (LR), Decision Tree (DT), and K-Nearest Neighbors (KNN).

The dataset was split into 80% training and 20% test sets using stratified sampling to maintain label balance. Accuracy, precision, recall, and F1-score were used to assess model performance, with particular attention to model generalization across both classes (Control vs. Patient).

A. Logistic Regression (LR)

Logistic Regression achieved the best performance among the three models with an overall accuracy of **83.7%**. LR yielded balanced class-wise precision (0.85 for Control, 0.82 for Patient) and recall (0.82 and 0.86 respectively), resulting in an overall F1-score of 0.84. This suggests a strong linear separability in the feature space.

B. Decision Tree (DT)

The Decision Tree model yielded a lower accuracy of **72.5%**. While easy to interpret, DTs tend to overfit on small feature spaces and had limited generalization. The confusion matrix showed a bias in misclassifying PD samples as Control (35 false negatives).

C. K-Nearest Neighbors (KNN)

The KNN model ($k = 5$) attained an accuracy of **81.4%**, with a more balanced confusion matrix. It correctly classified 121 out of 146 PD samples. However, it still misclassified 35 Control samples, indicating a tendency to lean toward the denser PD feature region in some cases.

D. Comparative Analysis and Interpretation

A comparison across all three models is shown in Table II.

TABLE II
SUMMARY OF MODEL PERFORMANCE

Model	Accuracy	Precision	F1 Score
Logistic Regression	83.7%	0.84	0.84
Decision Tree	72.5%	0.73	0.73
K-Nearest Neighbors	81.4%	0.81	0.81

Logistic Regression emerged as the most reliable model, benefiting from the linear boundaries defined by statistically significant features (RD and SD). Decision Trees, though interpretable, overfit and performed inconsistently. KNN displayed good generalization but was sensitive to local feature densities.

These results validate the importance of combining statistical testing (t-tests) and PCA with traditional models to uncover feature efficacy. The methods build an interpretable baseline for classification of Parkinsonian handwriting patterns that can be expanded on and compared against more complex machine learning methods such as Vision Transformers.

VI. VISION TRANSFORMER (ViT) ANALYSIS

A. Motivation

While traditional models provided interpretable and fairly accurate baselines, we aimed to evaluate whether more expressive and scalable deep learning models could improve performance. In particular, we investigated the use of Vision Transformers (ViTs), which are designed to capture global context in image data and have shown strong results in various visual recognition tasks. Much like a CNN, the ViT is used to classify image sets; however the method utilized differs. CNNs operate by focusing on features of an image, identified through multiple convolution and pooling operations. A ViT utilizes attention modules that can learn relationships between patches of an entire input image. The ability to identify relationships over a continuous image makes them particularly suited for detecting the nuanced motor irregularities in Parkinson's Disease (PD) handwriting patterns, such as tremors, stroke instability, and structural deviations. We also wanted to explore a relatively newer machine learning method for this task.

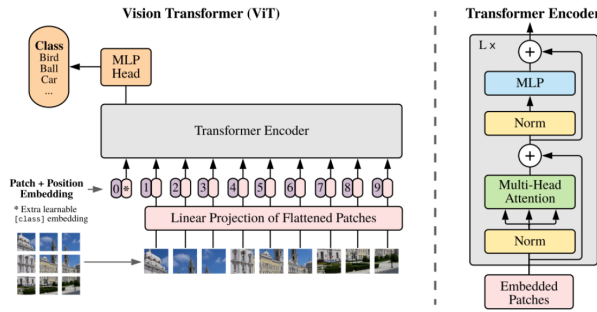


Fig. 7. ViT Architecture from An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale [6]

B. Analysis

In our analysis, we utilized the model introduced in “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”, pre-trained on the ImageNet21k which contains a large sample size of generic images [6]. This model was the foundation for ViTs, and can be used as the pre-trained model as well as be re-trained on our data. Using the pre-trained model, even if not trained on our data, we could analyze how transfer learning from generalized data might compare to a trained model. Afterward, we could train the model on our Parkinson handwriting data and compare results. Another comparison made was the accuracy of the model before and after the CycleGAN images were added. For each model, we separated the control and Parkinson patient images into training, validation, and test sets. This way we could make sure there was no leakage of the input information. We trained the model for 50 epochs and also implemented early stopping when the validation accuracy did not improve over 5 epochs to avoid overfitting of the training data. Figure 8 shows a confusion matrix from our ViT model trained on our HandPD and CycleGAN image set. We then collected the model diagnostic information shown in Table III.

TABLE III
PERFORMANCE OF ViT MODELS

Model	Accuracy	Precision	Recall	F1 Score
Pre-Trained (Only HandPD)	75.00%	86.29%	84.25%	85.26%
Pre-Trained (w/ CycleGAN)	54.24%	77.07%	55.00%	64.19%
Trained (Only HandPD)	91.89%	98.39%	92.42%	95.31%
Trained (w/ CycleGAN)	89.15%	88.54%	90.85%	89.68%

C. Results

Looking at the results in our table, the pre-trained models performed significantly worse in every metric compared to the model trained on our data, from. In accuracy, moving from 75% and 54.24%, up to 91.89% and 89.15% for data

without and with CycleGAN respectively. Transfer learning can be important when translating predictions from similar domains, however in this case it was not similar enough. The model trained on relevant data will be able to identify differences better than one that has not. The model metrics also decreased slightly when comparing the one that did not use the CycleGAN images to the one that did. The precision in the trained set that only has HandPD data was up to 98.39%, most likely because of the dataset imbalance. The model would have less chance to misplace a control if there were fewer of them. With the better balance between the control and patient groups, the model cannot overfit to a smaller sample size of majority patient information. We also observed that some misclassifications in the ViT model tended to occur with ambiguous handwriting patterns; especially cases where the tremor-induced distortions in patient samples were subtle, or when control samples exhibited slight irregularities. This suggests that while ViTs are robust, edge cases with overlapping feature characteristics may still pose challenges.

Continuing our overall analysis, we will utilize the model trained on the HandPD data including the CycleGAN generated images. The accuracy with this model increases when compared to the transfer learning models, and allows for training on a larger, even group, diverse dataset.

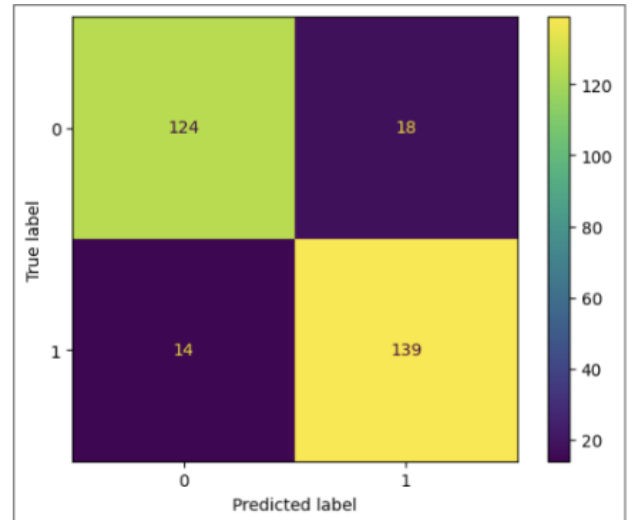


Fig. 8. Confusion matrix for ViT model with CycleGAN images (0: Control, 1: Patient)

VII. CONCLUSION

The objective of this study was to explore the effectiveness of modern machine learning techniques, particularly Vision Transformers (ViTs), in identifying Parkinson’s Disease (PD) from spiral handwriting samples—a task traditionally approached using shallow statistical models or CNNs. The central research question was:

Do Vision Transformers (ViTs) perform better than traditional methods such as logistic regression, decision tree, and

K nearest neighbors in predicting whether a given patient has Parkinson's based on traced templates that observe tremors?

To address this, we implemented a comprehensive pipeline involving both handcrafted feature extraction and deep learning-based modeling.

A. Comparative Performance Summary

We evaluated both traditional machine learning classifiers (Logistic Regression, Decision Tree, KNN) on handcrafted features, and a custom Vision Transformer trained on raw images (real + CycleGAN augmented). Our findings revealed the following:

- **Handcrafted Features + Logistic Regression:** Achieved an accuracy of **83.7%**. Radial Deviation and Stroke Density were statistically significant in separating Parkinson's disease from control samples (p -values < 0.05). This method provided interpretability but limited expressiveness for complex spatial variations.
- **Vision Transformer (ViT):** Outperformed traditional models with an accuracy of **89.15%**. The ViT model was able to learn rich, spatially-aware representations from raw spiral images and showed robustness against noise and irregular stroke patterns introduced by Parkinson's disease.
- **CNN Baselines (referenced in prior literature):** Typically range from 82–88% accuracy on similar datasets, often requiring more extensive preprocessing and data augmentation. However, CNNs are limited by local receptive fields, whereas ViTs leverage global attention mechanisms.

B. Findings

Our findings clearly indicate that Vision Transformers (ViTs) offer a powerful and scalable solution to the task of PD detection via handwriting analysis. Compared to handcrafted feature models, ViTs eliminate the need for manual feature engineering and can extract both local and global patterns of tremor-induced distortions.

The use of CycleGAN-based image synthesis further balanced the dataset and enhanced the generalizability of ViTs, proving that data augmentation is a key enabler in medical AI pipelines with limited samples.

C. Final Remarks

In conclusion, while traditional models like Logistic Regression offer interpretable baselines, modern architectures such as ViTs provide superior performance, adaptability, and potential for real-world deployment. The combination of domain-aware augmentation, handcrafted feature validation, and transformer-based modeling sets a robust foundation for future work in AI-assisted neurological diagnostics. Given the lightweight nature of our models and the use of readily collectible handwriting data, this approach could potentially be integrated into low-cost, non-invasive screening tools in clinical or telemedicine settings to support early-stage Parkinson's detection.

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